

Technical Data

Product Description

EXL9330 is an opaque (PC)-siloxane copolymer resin that offers extreme low temperature (-60 C) ductility in combination with excellent processability. This UV stabilized, medium flow resin features UL f1/V-0/5VA rating with non-chlorinated, non-brominated flame retardant agents for wide range of colors. EXL9330 is an excellent candidate for a broad range of applications such as mobile phones, tablets, industrial housings, electric circuit protection, personal safety helmets, electric vehicle supply equipment (EVSE) housings and connectors.

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E207780-228378 • E207780-102516601		
Search for UL Yellow Card	• SABIC		
Availability	• Asia Pacific		
Features	• Bromine Free • Chlorine Free	• Flame Retardant • Halogen Free	• High Impact Resistance • Low Temperature Impact Resistance
Uses	• Aerospace Applications • Appliances • Automotive Exterior Parts • Automotive Under the Hood • Construction Applications • Decorative Parts • Electric Vehicle Applications • Electrical Parts • Electrical/Electronic Applications	• Electronic Displays • Energy Storage • Fluid Handling • Industrial Applications • Lawn and Garden Equipment • Lighting Applications • Material Handling • Medical Devices • Medical/Healthcare Applications	• Military/Defense Applications • Optical Applications • Outdoor Applications • Personal Care • Rail Applications • Recreational Vehicle Applications • Sporting Goods • Water Management
Processing Method	• Injection Molding		
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value Unit	Test Method
Density / Specific Gravity		
--	1.18 g/cm ³	ASTM D792
--	1.19 g/cm ³	ISO 1183
Melt Mass-Flow Rate (MFR) (300°C/1.2 kg)	10 g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (300°C/1.2 kg)	9.0 cm ³ /10min	ISO 1133
Molding Shrinkage		Internal Method
Across Flow : 3.20 mm	0.40 to 0.80 %	
Flow : 3.20 mm	0.40 to 0.80 %	
Water Absorption		ISO 62
Saturation, 23°C	0.35 %	
Equilibrium, 23°C, 50% RH	0.15 %	
Outdoor Suitability	f1	UL 746C
Mechanical	Nominal Value Unit	Test Method
Tensile Modulus		
-- ³	2100 MPa	ASTM D638
--	2100 MPa	ISO 527-1/1
Tensile Strength		
Yield ⁴	58.0 MPa	ASTM D638
Yield	55.0 MPa	ISO 527-2/50
Break ⁴	61.0 MPa	ASTM D638
Break	60.0 MPa	ISO 527-2/50
Tensile Elongation		
Yield ⁴	6.0 %	ASTM D638
Yield	6.0 %	ISO 527-2/50
Break ⁴	130 %	ASTM D638
Break	130 %	ISO 527-2/50



LEXAN™ Copolymer EXL9330 - Asia

Polycarbonate

SABIC**PROSPECTOR®**

www.ulprospector.com

Mechanical	Nominal Value Unit	Test Method
Flexural Modulus		
50.0 mm Span ⁵	2060 MPa	ASTM D790
-- ⁶	2200 MPa	ISO 178
Flexural Stress		
-- ^{6, 7}	85.0 MPa	ISO 178
Yield, 50.0 mm Span ⁵	88.0 MPa	ASTM D790
Impact	Nominal Value Unit	Test Method
Charpy Notched Impact Strength ⁸		ISO 179/1eA
-30°C	60 kJ/m ²	
23°C	75 kJ/m ²	
Charpy Unnotched Impact Strength ⁸		ISO 179/1eU
-30°C	No Break	
23°C	No Break	
Notched Izod Impact		
-50°C	590 J/m	ASTM D256
-30°C	680 J/m	ASTM D256
23°C	800 J/m	ASTM D256
23°C ⁹	1100 J/m	Internal Method
23°C, 6.40 mm	640 J/m	ASTM D256
-30°C ¹⁰	55 kJ/m ²	ISO 180/1A
-30°C ¹¹	65 kJ/m ²	ISO 180/4A
23°C ¹⁰	70 kJ/m ²	ISO 180/1A
23°C ¹¹	80 kJ/m ²	ISO 180/4A
Unnotched Izod Impact Strength ¹⁰		ISO 180/1U
-30°C	No Break	
23°C	No Break	
Instrumented Dart Impact		ASTM D3763
23°C, Total Energy	52.0 J	
Hardness	Nominal Value Unit	Test Method
Ball Indentation Hardness (H 358/30)	90.0 MPa	ISO 2039-1
Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		
0.45 MPa, Unannealed, 3.20 mm	134 °C	ASTM D648
0.45 MPa, Unannealed, 4.00 mm, 100 mm Span ¹²	135 °C	ISO 75-2/Be
1.8 MPa, Unannealed, 3.20 mm	120 °C	ASTM D648
1.8 MPa, Unannealed, 6.40 mm	124 °C	ASTM D648
1.8 MPa, Unannealed, 4.00 mm, 100 mm Span ¹²	124 °C	ISO 75-2/Ae
Vicat Softening Temperature		
--	142 °C	ASTM D1525 ¹³ ISO 306/B120 ¹³
--	140 °C	ISO 306/B50
Ball Pressure Test (123 to 127°C)	Pass	IEC 60695-10-2
CLTE		
Flow : -40 to 40°C	6.7E-5 cm/cm/°C	ASTM E831
Flow : 23 to 80°C	7.2E-5 cm/cm/°C	ISO 11359-2
Transverse : -40 to 40°C	6.7E-5 cm/cm/°C	ASTM E831
Transverse : 23 to 80°C	7.7E-5 cm/cm/°C	ISO 11359-2
RTI Elec	125 °C	UL 746B
RTI Imp	115 °C	UL 746B
RTI Str	125 °C	UL 746B



LEXAN™ Copolymer EXL9330 - Asia

Polycarbonate

SABIC**PROSPECTOR®**

www.ulprospector.com

Electrical	Nominal Value Unit	Test Method
Surface Resistivity	> 1.0E+15 ohms	IEC 60093
Volume Resistivity	> 1.0E+15 ohms·cm	IEC 60093
Dielectric Strength		
3.20 mm, in Oil	17 kV/mm	ASTM D149
3.20 mm, in Oil	16 kV/mm	IEC 60243-1
Dielectric Constant		
60 Hz	2.95	ASTM D150
50 kHz	2.95	ASTM D150
1 MHz	2.90	ASTM D150
50 Hz	2.60	IEC 60250
60 Hz	2.60	IEC 60250
1 MHz	2.70	IEC 60250
Dissipation Factor		
50 Hz	2.4E-3	ASTM D150
60 Hz	2.4E-3	ASTM D150
1 MHz	8.5E-3	ASTM D150 IEC 60250
50 Hz	1.0E-3	IEC 60250
60 Hz	1.0E-3	IEC 60250
Comparative Tracking Index (CTI)	PLC 3	UL 746A
Comparative Tracking Index	225 V	IEC 60112
High Amp Arc Ignition (HAI)		UL 746A
> 0.60 mm	PLC 1	
> 2.3 mm	PLC 0	
Hot-wire Ignition (HWI)		UL 746A
> 0.60 mm	PLC 3	
> 1.5 mm	PLC 2	
> 3.0 mm	PLC 1	
Flammability	Nominal Value Unit	Test Method
Flame Rating		UL 94
> 0.60 mm	HB	
> 0.8 mm	V-1	
> 1.5 mm	V-0	
> 2.5 mm	5VB	
> 3.0 mm	5VA	
Glow Wire Flammability Index		IEC 60695-2-12
1.0 mm	960 °C	
1.5 mm	960 °C	
2.0 mm	960 °C	
2.3 mm	960 °C	
2.5 mm	960 °C	
3.0 mm	960 °C	
Glow Wire Ignition Temperature		IEC 60695-2-13
1.0 mm	825 °C	
1.5 mm	825 °C	
2.0 mm	825 °C	
2.3 mm	825 °C	
2.5 mm	825 °C	
3.0 mm	825 °C	
Oxygen Index	35 %	ISO 4589-2
Injection	Nominal Value Unit	
Drying Temperature	120 °C	



Injection	Nominal Value Unit
Drying Time	3.0 to 4.0 hr
Suggested Max Moisture	0.020 %
Suggested Shot Size	40 to 60 %
Rear Temperature	235 to 295 °C
Middle Temperature	245 to 305 °C
Front Temperature	255 to 315 °C
Nozzle Temperature	250 to 310 °C
Processing (Melt) Temp	255 to 315 °C
Mold Temperature	70 to 95 °C
Back Pressure	0.300 to 0.700 MPa
Screw Speed	40 to 70 rpm
Vent Depth	0.025 to 0.076 mm

Injection Notes

- Drying Time (Cumulative): 48 hr

Notes

¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.

² Typical properties: these are not to be construed as specifications.

³ 50 mm/min

⁴ Type I, 50 mm/min

⁵ 1.3 mm/min

⁶ 2.0 mm/min

⁷ at Yield

⁸ 80*10*3 sp=62mm

⁹ Double-gated

¹⁰ 80*10*3 mm

¹¹ 63.5*12.7*3.2 mm

¹² 120*10*4 mm

¹³ Rate A (50°C/h), Loading 2 (50 N)



LEXAN™ Copolymer EXL9330 - Asia

Polycarbonate

SABIC

PROSPECTOR®

www.ulprospector.com

Where to Buy

Supplier

SABIC

Web: <http://www.sabic.com/>

Distributor

Please contact the supplier to find a distributor for LEXAN™ Copolymer EXL9330 - Asia

